

Problem 2

$$\mathbb{P}(1 \text{ wins on first round}) = \frac{6}{36} = \frac{1}{6}$$

$$\mathbb{P}(2 \text{ wins on first round}) = \frac{1}{36} + \frac{2}{36} + \frac{3}{36} = \frac{1}{6}$$

$$\mathbb{P}(\text{tie on first round}) = \mathbb{P}(1 \text{ loses on first round})\mathbb{P}(2 \text{ loses on first round}) = \left(\frac{5}{6}\right)^2 = \frac{25}{36}$$

$$\mathbb{P}(1 \text{ wins}) = \mathbb{P}(1 \text{ wins round}) + \mathbb{P}(\text{tie})\mathbb{P}(1 \text{ wins round}) + \mathbb{P}(\text{ties2})\mathbb{P}(1 \text{ wins round}) + \dots$$

$$\mathbb{P}(1 \text{ wins}) = \frac{1}{6} + \frac{1}{6} \frac{25}{36} + \frac{1}{6} \left(\frac{25}{36}\right)^2 + \dots$$

$$\mathbb{P}(1 \text{ wins}) = \frac{\frac{1}{6}}{1 - \frac{25}{36}} = \frac{6}{11}$$

Problem 6

1. There is a total of $13!$ permutations to order 13 cards.

To count the ways that the Jack, King, and Queen are next to each other we can treat them as 1 card, so there is $3!$ ways to order those cards and $11!$ ways to order the 10 cards plus our group of 3 thus

$$\mathbb{P}(E) = \frac{3! \times 11!}{13!}$$

2. Again there is a total of $13!$ permutations to order 13 cards.

There are 3 ways to pair each element KQ, KJ, QJ and using a proof similar to in 1. we know

$$\mathbb{P}(KQ) = \frac{2! \times 12!}{13!}, \quad \mathbb{P}(KJ) = \frac{2! \times 12!}{13!}, \quad \mathbb{P}(QJ) = \frac{2! \times 12!}{13!}$$

$\mathbb{P}(KQ \cap KJ) \implies$ we either have QKJ or JKQ thus

$$\mathbb{P}(KQ \cap KJ) = \frac{2! \times 11!}{13!}, \quad \mathbb{P}(KQ \cap QJ) = \frac{2! \times 11!}{13!}, \quad \mathbb{P}(QJ \cap KJ) = \frac{2! \times 11!}{13!}$$

$\mathbb{P}(KQ \cap KJ \cap QJ) = 0$ since we can't make them all adjacent to each other.

Thus, by inclusion exclusion

$$\mathbb{P}(E) = 1 - \mathbb{P}(KQ) - \mathbb{P}(KJ) - \mathbb{P}(QJ) + \mathbb{P}(KQ \cap KJ) + \mathbb{P}(KQ \cap QJ) + \mathbb{P}(QJ \cap KJ) - \mathbb{P}(KQ \cap KJ \cap QJ)$$

$$\mathbb{P}(E) = 1 - 3 \frac{2! \times 12!}{13!} + 3 \frac{2! \times 11!}{13!}$$

3. Again there is a total of $13!$ permutations to order 13 cards.

The number of ways to count this is first we start with $13!$ but since there is $3!$ ways to order the K,Q,J and we only care about one we over count by a factor of $3!$ thus the number of

ways is $13!/3!$ thus $\mathbb{P}(E) = \frac{\frac{13!}{3!}}{13!} = \frac{1}{6}$

Problem 7

1. LHS: This is the number of ways to pick n objects from $2n$ objects

RHS: We first split our $2n$ objects into sets of n objects we then take k objects from the first $\binom{n}{k}$ and take $n - k$ from the second $\binom{n}{k} = \binom{n}{n-k}$ we then sum from 0 to n to get all combinations of picking from set 1 and set 2 thus the sum is equivalent to counting the number of ways to pick n objects from $2n$ objects

2. LHS: This is the number of ways to make m long words with 3 letter alphabet lets say a, b, c

RHS: We first set all letters to a we then pick k letters from our m $\binom{m}{k}$ then we pick either b or c for those k letters 2^k . We then sum from 0 to m to get all combinations of picking from the letter a 's thus the sum is equivalent to counting the number of ways to make m long words with 3 letter alphabet

Problem 9

$$1. \mathbb{P}(\text{first head on the } n\text{-th throw}) = \mathbb{P}(n-1 \text{ tails})\mathbb{P}(\text{heads}) = \left(\frac{1}{2}\right)^{n-1} \times \frac{1}{2} = \left(\frac{1}{2}\right)^n$$

$$2. \mathbb{P}(\# \text{ of heads and tails are equal on } n\text{-th throw})$$

Total number of possibilities is 2^n

If n is odd $\mathbb{P}(\# \text{ of heads and tails are equal on } n\text{-th throw}) = 0$

If n is even let right be a heads and let up be a tails if we want there to be an equal number of heads and tails then we want $n/2$ heads and $n/2$ tails thus this counts the number of lattice paths from $0,0$ to $n/2, n/2$ thus this is $\binom{n}{n/2}$ thus

$$\mathbb{P}(\# \text{ of heads and tails are equal on } n\text{-th throw}) = \frac{\binom{n}{n/2}}{2^n}$$

$$3. \mathbb{P}(\text{exactly two heads have appeared to date})$$

Total number of possibilities is 2^n

The number of ways we get exactly two heads is just $\binom{n}{2}$ since we are picking 2 heads and the rest are tails thus

$$\mathbb{P}(\text{exactly two heads have appeared to date}) = \frac{\binom{n}{2}}{2^n}$$

$$4. \mathbb{P}(\text{at least two heads have appeared to date})$$

Total number of possibilities is 2^n

The number of ways to get at least two heads is two first pick two heads from our n coin flips which is $\binom{n}{2}$ then the remaining $n-2$ flips have 2 choices thus 2^{n-2} so the total number of ways is $\binom{n}{2} \times 2^{n-2}$ thus

$$\mathbb{P}(\text{at least two heads have appeared to date}) = \frac{\binom{n}{2} \times 2^{n-2}}{2^n}$$